

Work Breakdown Structure (WBS)

Municipal Infrastructure Damage Reporting App – Problem Statement #29

WBS ID	Phase	Task	Owner	Duration
1.0	Requirements Engineering	Scenario review and stakeholder identification (Citizen Reporter, Municipal Engineer)	Analyst	1 day
1.1	Requirements Engineering	Elicit and draft functional requirements (FR-001 to FR-005)	Analyst	1 day
1.2	Requirements Engineering	Elicit and draft nonfunctional requirements (NFR-001, NFR-002)	Analyst	0.5 day
1.3	Requirements Engineering	Peer review / critique of requirements for clarity and testability	Team	0.5 day
1.4	Requirements Engineering	Finalize Requirements Table with acceptance criteria and rationale	Analyst	0.5 day
2.0	System Design	Identify actors and use cases from approved requirements	Designer	0.5 day
2.1	System Design	Model UML use-case diagram (actors, use cases, include/extend)	Designer	1 day
2.2	System Design	Draft use-case flow specification (main scenario + alternate flow)	Designer	1 day
2.3	System Design	Design system architecture (client, service, data, external layers)	Architect	1 day
2.4	System Design	Design database schema (reports, tickets, department directory)	Architect	1 day
3.0	Development	Build Citizen Mobile App (report submission, photo upload, status tracking)	Dev – Mobile	5 days
3.1	Development	Build Municipal Engineer Web Portal (ticket queue, status updates)	Dev – Web	4 days
3.2	Development	Implement API Gateway and Report Submission Service	Dev – Backend	3 days
3.3	Development	Implement GPS/EXIF Extraction Module and GIS integration	Dev – Backend	2 days
3.4	Development	Implement Ticket Routing & Dispatch Engine	Dev – Backend	2 days
3.5	Development	Implement Notification Service (SMS/email/push integration)	Dev – Backend	2 days

WBS ID	Phase	Task	Owner	Duration
3.6	Development	Implement Offline Sync Manager	Dev – Mobile	2 days
4.0	Testing & QA	Write unit tests for core services (submission, routing, extraction)	QA	2 days
4.1	Testing & QA	Integration testing across mobile app, backend, and GIS system	QA	2 days
4.2	Testing & QA	Acceptance testing against FR/NFR acceptance criteria	QA	1.5 days
4.3	Testing & QA	Security and load testing (NFR-001, NFR-002)	QA	1.5 days
5.0	Deployment & Documentation	Deploy backend services and mobile app to staging environment	DevOps	1 day
5.1	Deployment & Documentation	Prepare SRS, architecture, and use-case documentation for submission	Analyst	1 day
5.2	Deployment & Documentation	Push final deliverables to GitHub repository	Team	0.5 day
5.3	Deployment & Documentation	Production rollout and handover to municipal IT	DevOps	1 day

Estimated total effort: ~35.5 working days across phases (durations are per-task estimates; team members work in parallel within a phase, so phase duration is shorter than the sum of its task durations).